25 A flat circular coil of 120 turns, each of area 0.070 m^2, is placed with its axis parallel to a uniform magnetic field. The flux density of the field is changed steadily from 80 mT to 20 mT over a period of 4.0 s.

What is the e.m.f. induced in the coil during this time?

A 0 mV **B** 1.1 mV **C** 130 mV **D** 500 mV

Ans: C

Axis of coil parallel to magnetic field → magnetic field lines are normal to the area enclosed by coil

Emf = $\frac{d\phi}{dt}$ = $NA\frac{dB}{dt} = 120 \times 0.070 \times \frac{(80-20)\times 10^{-3}}{4.0}$

$= 0.126\,\text{V} \approx 130\,\text{mV}$

26 The figure belows shows a varying voltage